FINAL EXPLANATION: CHEMISTRY GRADE 10

Student Assessment Document

FINAL EXPLANATION: CHEMISTRY | GRADE 10 | SUB-STRAND 1.4: CHEMICAL BONDING	
Student Assessment Document	
Student Name	______________________________
Class	______________________________
Date	______________________________

INSTRUCTIONS FOR STUDENTS

This is your Final Explanation task. Over the past 13 lessons, you have investigated one of chemistry's most fascinating puzzles — the Carbon Paradox. Now it is time to pull everything you have learned together into one well-reasoned, evidence-based explanation. Read the driving question carefully, then write your response section by section. Use scientific vocabulary you have learned, refer back to evidence from your investigations, and connect every idea to the anchoring phenomenon of diamond and graphite. Your explanation should show not just WHAT you know, but WHY and HOW — that is the heart of competency-based learning.

Section 1: Valence Electrons and the Type of Bond Formed	
Explain what valence electrons are and how the number of valence electrons a carbon atom has determines the type of chemical bond it can form. Why does carbon form covalent bonds rather than ionic bonds? Use the carbon atom's electron configuration as evidence, and connect your answer to the diamond and graphite phenomenon.	Carbon is element number 6 on the periodic table. Its electron configuration is 2, 4 — meaning it has 4 valence electrons in its outermost energy level. To achieve a stable, full outer shell of 8 electrons (the octet rule), a carbon atom needs to gain or share 4 more electrons. Because carbon sits in the middle of Period 2, it neither strongly attracts nor strongly donates electrons, so it does not form ions easily. Instead, carbon achieves stability by SHARING electrons with neighbouring atoms — this is called covalent bonding. In a covalent bond, two atoms each contribute one electron to form a shared pair that 'belongs' to both atoms simultaneously. Because carbon has 4 valence electrons available for sharing, it can form up to 4 covalent bonds at the same time. This is the key to understanding the Carbon Paradox: in both diamond and graphite, every carbon atom forms covalent bonds using those same 4 valence electrons — yet the NUMBER of bonds each carbon forms (4 in diamond, 3 in graphite) and the DIRECTION those bonds point in space is what makes the two substances completely different.

Section 2: The Atomic Structure of Diamond and Graphite	
Describe the bonding structure inside diamond and inside graphite. For each substance: state how many covalent bonds	In diamond, each carbon atom uses all 4 of its valence electrons to form 4 strong covalent bonds with 4 neighbouring carbon atoms. These 4 bonds point outward in four different directions, equally spaced in three-dimensional space — this shape is called a tetrahedral arrangement (bond angles of approximately 109.5°). Every single carbon atom in the entire crystal is bonded this way, creating one giant, continuous three-dimensional covalent network that extends in all directions. There are no free electrons left

each carbon atom forms, describe the shape or arrangement of those bonds in space, and explain what happens to any valence electrons that are NOT used in bonding. Draw or describe a labelled diagram of each structure if it helps your explanation.	over, because all 4 valence electrons are locked into bonds. The result is an enormous, rigid, interlocking lattice — like a microscopic three-dimensional chain-link structure that has no weak points. In graphite, each carbon atom uses only 3 of its 4 valence electrons to form 3 covalent bonds with 3 neighbouring carbon atoms. These 3 bonds lie flat in one plane and point outward at 120° angles from each other, creating a hexagonal (six-sided) ring pattern — like a honeycomb tile — that extends outward in two dimensions to form a flat sheet called a graphene layer. Millions of these hexagonal sheets are stacked on top of each other. Crucially, the 4th valence electron of each carbon atom is NOT used in a bond within the layer. Instead, these leftover electrons are delocalised — they are free to move across the entire layer and are not 'owned' by any single atom. Between the layers, there are only weak intermolecular forces (van der Waals forces), not covalent bonds.

Section 3: How Bonding Structure Determines Physical Properties	
Use the bonding structures you described in Section 2 to explain at least THREE contrasting physical properties of diamond and graphite: hardness, electrical conductivity, and one other property of your choice (e.g. texture/feel, melting point, opacity, or thermal conductivity). For each property, your explanation must link directly to the bonding structure — do not just state the property, explain the structural reason for it.	HARDNESS: Diamond is the hardest naturally occurring substance known — it scored 10 on the Mohs scale and scratched the ceramic tile in our classroom investigation. This extreme hardness exists because every carbon atom is locked into 4 strong covalent bonds in all three dimensions. To scratch or deform diamond, you would have to break millions of these strong covalent bonds simultaneously — this requires an enormous amount of energy, so virtually no material can do it. Graphite, by contrast, is so soft it leaves a grey smear on paper (that is how pencils work!). The carbon atoms within each graphene layer are held together strongly, but the layers themselves are only held to each other by weak van der Waals forces. When you press a pencil on paper, these weak forces are overcome easily and layers slide off onto the page. The structure explains everything: 3-D network of strong bonds = extreme hardness; 2-D layers with weak interlayer forces = softness and slipperiness. ELECTRICAL CONDUCTIVITY: In our investigation, graphite lit the conductivity bulb brightly while diamond did not conduct at all. Electrical conduction requires charged particles — usually electrons — that are free to move through the material. In graphite, the delocalised 4th electrons from every carbon atom form a 'sea' of mobile electrons spread across each graphene layer. When a voltage is applied, these electrons flow, carrying electric current — just like the free electrons in a metal. In diamond, ALL 4 valence electrons of every carbon atom are locked tightly into covalent bonds and cannot move freely. There are no mobile charge carriers, so diamond cannot conduct electricity and is used as an electrical insulator. MELTING POINT: Both diamond and graphite have very high melting points compared to most substances — diamond melts above 3 500 °C, and graphite sublimes at around 3 600 °C. This is because in BOTH structures, the carbon atoms within the lattice (diamond) or within each layer (graphite) are connected by strong covalent bonds. To melt either substance, you must supply enough energy to break these strong covalent bonds, which requires extremely high temperatures. This contrasts sharply with ionic compounds like table salt ($NaCl$, melting point 801 °C) or molecular substances like water (0 °C), where only weaker forces need to be overcome. The high melting point is direct evidence of strong covalent bonding in both allotropes.

Section 4: From Structure to Real-World Uses	
Explain how the different bonding structures of diamond and graphite make each substance suited to specific real-world uses. Give at least	DIAMOND — USES EXPLAINED BY STRUCTURE: 1. Cutting and drilling tools: The Nairobi jeweller's cutting tool shown at the start of our unit uses a diamond-tipped blade. Because diamond is the hardest natural substance (arising from its 3-D tetrahedral covalent network), it can cut, grind, or polish materials that no other natural substance can scratch. Industrial diamond-tipped drill bits are also used in mining operations across Africa,

TWO uses for each substance and explain, for each use, which specific property (arising from the bonding structure) makes it suitable. Where possible, refer to Kenyan or African contexts.	including in Tanzania's gemstone mines and South African gold mines, to drill through extremely hard rock. 2. Gemstones and jewellery: Diamond's 3-D bonded structure causes light to refract and reflect in a dazzling way, making it transparent and brilliant. Its hardness also means it does not scratch or wear away over time, so it retains its polished surface permanently — ideal for jewellery that is worn every day. GRAPHITE — USES EXPLAINED BY STRUCTURE: 1. Pencil 'lead': The graphite in the pencil demonstrated at the start of our unit writes on paper because the weak van der Waals forces between graphene layers allow thin layers to slide off onto the paper surface easily, leaving a visible grey mark. The strong bonds within each layer keep the graphite solid enough to be held in a pencil casing. 2. Electrodes in electrolysis: Because graphite conducts electricity (due to its delocalised electrons) and can withstand very high temperatures (due to the strong in-layer covalent bonds), it is used as an electrode in industrial electrolysis. For example, in East Africa, graphite electrodes are used in electrolytic refining of metals and in the production of chlorine from brine. Graphite is also used as a lubricant in machinery — the layers slide over each other easily, reducing friction between moving metal parts.

Section 5: Answering the Driving Question	
Now bring everything together. Write a full, coherent answer to the driving question: 'Why do diamond and graphite — both made of only carbon atoms — have such completely different properties, and how does the way atoms bond together determine what a substance looks like, how it behaves, and what we can use it for?' Your answer should synthesise ideas from all previous sections, use correct scientific terminology, and refer back to at least ONE piece of evidence from the classroom investigation. Aim for 3–5 paragraphs.	Diamond and graphite are both made of 100% carbon atoms — the same element, with the same number of protons, neutrons, and electrons. Yet as we saw at the start of this unit, they could not behave more differently: diamond scratched the ceramic tile and refused to conduct electricity, while graphite smeared a grey mark and lit the conductivity bulb. The answer to this paradox lies entirely in how the carbon atoms are bonded to one another — not in WHAT the atoms are, but in the ARCHITECTURE of their bonds. Every carbon atom has 4 valence electrons, which it shares through covalent bonds to achieve a stable octet. In diamond, each carbon forms 4 covalent bonds pointing outward in a tetrahedral arrangement in all three dimensions, creating one enormous, continuous covalent network with no free electrons. This rigid 3-D lattice makes diamond extraordinarily hard (hardest natural substance on Earth), a perfect electrical insulator (no free electrons to carry current), and gives it a very high melting point (millions of strong bonds must all be broken). In graphite, each carbon forms only 3 covalent bonds, all lying flat in a hexagonal layer. The 4th valence electron is delocalised and free to move. These layers are stacked with only weak van der Waals forces between them. This layered structure makes graphite soft and slippery (layers slide apart easily), an excellent electrical conductor (delocalised electrons carry current), and opaque and grey in appearance. The different structures arise from the same atoms making different bonding choices — and those bonding choices produce substances with completely different properties. This is a fundamental principle of chemistry: the macroscopic properties we observe (hardness, conductivity, melting point, appearance) are determined by the microscopic structure of the bonds holding atoms together. Changing the bonding arrangement changes everything about how a substance behaves. This principle also explains why diamond and graphite have completely different uses, despite being the same element. Diamond's hardness and optical brilliance make it irreplaceable in cutting tools used by Nairobi jewellers and in African mining operations. Graphite's conductivity and lubricating layers make it useful in electrodes, pencils, and industrial machinery. Neither substance's use could be predicted just by knowing it is made of carbon — you have to know HOW those carbon atoms are bonded. In conclusion, the Carbon Paradox teaches us the most important lesson of this unit: chemical bonding is not just about which atoms join together, but about how they join, in what geometry, and with how many bonds. These structural decisions at the atomic

	scale ripple upward to determine every property we can see, measure, and use at the scale of the real world. This is the power of understanding chemical bonding.

RUBRIC			
Criterion	**Excellent (4)**	**Proficient (3)**	**Developing (1–2)**
1. Valence Electrons and Covalent Bonding	Accurately explains carbon's electron configuration (2,4) and all 4 valence electrons. Clearly and correctly explains why carbon forms covalent (not ionic) bonds using the octet rule. Explicitly links valence electron behaviour to both diamond and graphite bonding structures.	Correctly identifies carbon's 4 valence electrons and explains covalent bond formation. Connects valence electrons to the phenomenon but may lack full depth or precision in linking to both allotropes.	Shows partial understanding of valence electrons or covalent bonding. May confuse ionic and covalent bonding, or fail to connect the explanation to the carbon paradox phenomenon.
2. Accuracy of Structural Descriptions (Diamond and Graphite)	Provides a detailed, accurate description of both structures: tetrahedral 3-D network for diamond (all 4 bonds, ~109.5° bond angles); flat hexagonal layers for graphite (3 bonds, 120°, delocalised 4th electron, weak interlayer van der Waals forces). Uses correct scientific terminology throughout.	Describes both structures with mostly accurate details. May omit one structural feature (e.g. bond angles, or the nature of interlayer forces) or use imprecise terminology in one area.	Provides a basic or partially correct description of one or both structures. Key features such as delocalised electrons or the 3-D network may be missing, vague, or incorrect.
3. Linking Bonding Structure to Physical Properties	Explains at least THREE physical properties (hardness, electrical conductivity, and one other) for both diamond and graphite. Each explanation explicitly traces the property back to a specific structural feature (e.g. 'diamond cannot conduct because ALL 4 valence electrons are locked in bonds — there are no free charge carriers'). Evidence from the classroom investigation is referenced.	Explains at least TWO or THREE properties and links most of them to structural features. The connection between structure and property may be stated but not fully elaborated for all cases. Investigation evidence may be mentioned but not integrated deeply.	States properties of diamond and graphite but explanations are largely descriptive rather than structural. Little or no connection is made between bonding arrangement and the observed property. Investigation evidence is absent or incorrectly used.
4. Real-World Applications Linked to Properties	Gives at least TWO uses for each substance. For every use, clearly explains WHICH specific property makes it suitable AND traces that property back to the bonding structure. Incorporates at least one Kenyan or East African context accurately and relevantly.	Gives at least TWO uses for each substance with property explanations for most. The link back to bonding structure may be implicit rather than explicit in some cases. A Kenyan or African context is mentioned.	Lists uses but provides little or no explanation of why the substance is suited to those uses. The connection to bonding structure is absent or superficial. No Kenyan or African context is included.
5. Synthesis — Answering the Driving Question	The final response is a coherent, well-structured argument (3–5 paragraphs)	The final response addresses the driving question and connects most major ideas.	The final response attempts to answer the driving question but is largely a

	that integrates all unit concepts: valence electrons → bond type → bonding structure → properties → uses. Correct scientific terminology is used consistently. At least one piece of investigation evidence is cited. The response demonstrates genuine understanding of the principle that atomic-scale bonding determines macroscopic behaviour.	The argument is mostly coherent but may lack full integration of all concepts or have minor scientific inaccuracies. Terminology is mostly correct.	summary of facts rather than a synthesised argument. Connections between bonding, properties, and uses are weak or missing. Scientific terminology is limited or used incorrectly.